

Tanvir Ahmed Shanto

Mechanical Engineering Ph.D. Candidate

Experimental & Computational Solid Mechanics | Fracture Mechanics | Micromechanics FEA & Validation

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Research Profile

Mechanical Engineering Ph.D. candidate (defense expected August 2026) with more than seven years of experience in experimental and computational solid mechanics, fracture and damage mechanics, and additive manufacturing. Research integrates micromechanics and cohesive-zone FEA with ASTM-based testing, MWIR thermography, X-ray μ -CT, SEM fractography, and design of experiments to quantify process–structure–property–performance relationships in anisotropic polymer systems. Interests include multiscale modeling, experiment-to-model calibration and validation, coupled thermal–mechanical behavior, uncertainty quantification, Bayesian calibration, and data-driven surrogate modeling.

Research Experience

Graduate Student Researcher

The University of Texas at Arlington

Aug 2021–Present

Arlington, Texas, USA

- **Integrated** MWIR thermography, X-ray μ -CT, tensile/DMA/Mode-I testing, and SEM fractography; experimentally anchored linkage among interfacial thermal history, void morphology, and structural response
- **Designed and analyzed** a randomized $2 \times 3 \times 5$ full-factorial program (60 print batches); GLM/ANOVA, Type III sums of squares, interaction/residual diagnostics, and Tukey HSD; printhead–speed and sample-spacing effects resolved
- **Quantified** in-situ annealing gains in ABS-CF; 39% higher ultimate tensile strength, 30% higher elastic modulus, 63% higher glassy storage modulus, and approximately 92% higher Mode-I fracture resistance; crack-path voids reduced from approximately 4.0% to 0.7%; cohesive-fracture shift
- **Developed** μ -CT-informed Abaqus RVEs for standard and in-situ-annealed bead architectures; three-axis virtual homogenization, mesh convergence, and low/mid/high engineering-property envelopes
- **Constructed** Abaqus/Explicit DCB and SENB cohesive-zone models; one-factor studies of solid properties, normal cohesive strength T_{n0} , and Mode-I fracture energy G_{Ic} ; automated load–displacement, peak, stiffness, and fracture-work metrics
- **Performed** ASTM D5528-based DCB and ASTM D5045 SENB testing with synchronized force–displacement and crack tracking; calibration and validation targets for cohesive laws
- **Designed and prototyped** adaptive in-situ annealing printheads using GD&T and DFM/DFA; one issued and one pending U.S. patent; publication and commercialization impact

Education

Ph.D. in Mechanical Engineering

The University of Texas at Arlington

Sep 2021–Aug 2026

Arlington, Texas, USA

Dissertation: Optimizing Interlayer Adhesion in Fused Filament Fabrication (FFF) for Improved Mechanical Performance

Advisor: Dr. Robert M. Taylor **Defense expected:** August 2026

B.Sc. in Naval Architecture and Marine Engineering

Bangladesh University of Engineering and Technology (BUET)

Feb 2013–Sep 2017

Dhaka, Bangladesh

Thesis: Predicting crack propagation at vulnerable locations of ship structures and analyzing possible mitigation measures

Research and Technical Expertise

Computational FEA	Abaqus/Standard and Abaqus/Explicit, ANSYS, HyperWorks/HyperStudy, and CAESAR II; RVE homogenization, anisotropy, cohesive-zone/traction–separation modeling, contact and damage, mesh convergence, mass scaling, parameter sensitivity, ODB automation, and validation.
Experimental mechanics	ASTM-based tensile, compression, bending, fatigue, and DMA; DCB, SENB, and Mode-I fracture testing; synchronized crack tracking; load–displacement and energy reduction.
NDE & characterization	MWIR/IR thermography; X-ray μ -CT; optical and SEM/EDS fractography; DSC, TGA, STA, FTIR, XRD, and AFM.
Statistics, UQ & data	DoE; GLM/ANOVA; residual diagnostics; Tukey comparisons; sensitivity and calibration; Python, MATLAB, SAS Studio, Fortran; surrogate modeling.
Design & manufacturing	SolidWorks, AutoCAD/Plant 3D, ASME Y14.5 GD&T, DFM/DFA; FFF, SLA, SLS, material jetting, DLP, multi-axis extrusion, DFAM, and LPBF; ASTM and ISO methods.

Peer-Reviewed Journal Articles

1. **Shanto, T. A.**, Ahmed, R., Patel, P., Zulqernine, M. J., Chen, V., Jain, A., & Taylor, R. M. (2026). In-situ annealing and thermal-morphological evolution during extrusion-based additive manufacturing. *Journal of Manufacturing Processes*, 170, 303–318. doi:10.1016/j.jmapro.2026.05.005
2. Ahmed, R., Dola, I. S., Ahmed, S., Rahman, M. M., **Shanto, T. A.**, & Taylor, R. M. (2026). A review of design and fabrication strategies for thermoplastic lightweight thin-walled stiffened shells and lattice structures utilizing fused-filament fabrication (FFF). *International Journal of Advanced Manufacturing Technology*. doi:10.1007/s00170-026-18078-6
3. Patel, P., Ahmed, R., **Shanto, T. A.**, Jain, A., & Taylor, R. M. (2025). Experimental characterization of enhanced fused filament fabrication (FFF) of tall thin-walled structures using polylactic acid (PLA). *International Journal of Advanced Manufacturing Technology*, 139(11–12), 5663–5675. doi:10.1007/s00170-025-16171-w
4. Uddin, M. M., Akhtaruzzaman, R., **Shanto, T. A.**, & Hasan, M. N. (2023). Study the performance of a dynamic wall heat exchanger using computational fluid dynamics. *European Journal of Engineering and Technology Research*, 8(3), 38–42. doi:10.24018/ejeng.2023.8.3.3043
5. Patel, P., Rahman, M. M., **Shanto, T. A.**, Kulkarni, P., Ahmed, R., Jain, A., & Taylor, R. M. (2026). Optimization of in-process annealing for extrusion printing of tall thin-walled structures. *Progress in Additive Manufacturing*. In press.

Manuscripts Under Review

1. **Shanto, T. A.**, Rahman, M. M., Barua, S., Ahmed, S., Raiyan, A. Z., Ahmed, R., Jain, A., Adnan, A., & Taylor, R. M. In-situ annealing of vertically printed short-carbon-fiber ABS: A process–structure–property study of bonding potential, voids, and z-direction performance. Submitted to *Composites Part B: Engineering*.
2. **Shanto, T. A.**, Ahmed, R., Patel, P., Barua, S., Dola, I. S., Chen, V., Jain, A., & Taylor, R. M. Adaptive annealing in extrusion-based additive manufacturing and machine-learning-based prediction of mechanical performance. Manuscript under internal review.
3. Ahmed, R., **Shanto, T. A.**, Ahmed, S., Raiyan, A. Z., Adnan, A., Jain, A., & Taylor, R. M. Process–structure–property relationships in FFF PLA parts. Manuscript under internal review.

Works in Progress

1. **Shanto, T. A.**, Ahmed, S., Labanya, F. N., Lin, S., & Taylor, R. M. Characterizing and improving fracture behavior in 3D-printed polymers using DCB and SENB fracture testing. Ongoing research.
2. Rahman, M. M., **Shanto, T. A.**, Ahmed, R., & Taylor, R. M. Improving the sealing capacity of FFF ducts for aerospace environmental-control-system applications. Manuscript in preparation.
3. Ahmed, S., **Shanto, T. A.**, Ahmed, R., & Taylor, R. M. Improving the mechanical performance of lightweight PLA for unmanned aerial vehicles. Ongoing research.
4. Ahmed, R., Zulqernine, M. J., **Shanto, T. A.**, Lin, S., & Taylor, R. M. Improving the compressive strength of thin-walled stiffened cylindrical structures and characterizing damage using acoustic-emission sensing. Ongoing research.

Peer-Reviewed Conference Publications

1. Dola, I. S., Ahmed, R., Zulqernine, M. J., **Shanto, T. A.**, & Taylor, R. M. (2026). Evaluation of deep learning architectures for defect detection in fused filament fabrication. *SAMPE Conference and Exhibition*. doi:10.33599/nasampe/s.26.91
2. Ahmed, S., Rahman, M. M., **Shanto, T. A.**, Ahmed, R., & Taylor, R. M. (2026). A statistical approach for evaluating printing-temperature effects on lightweight and generic polylactic acid in fused filament fabrication. *IISE Annual Conference Proceedings*. In press.
3. Rahman, M. M., Ahmed, S., **Shanto, T. A.**, Ahmed, R., & Taylor, R. M. (2026). A statistical study on the influence of build-volume temperature and bead overlapping of PEI in fused filament fabrication. *IISE Annual Conference Proceedings*. In press.
4. Ahmed, R., **Shanto, T. A.**, Rahman, M. M., Taylor, R. M., & Jain, A. (2025). Machine learning-based prediction of toughness in fused filament fabrication: Leveraging in-process annealing with enhanced printheads. In *Solid Freeform Fabrication 2025: Proceedings of the 36th Annual International Solid Freeform Fabrication Symposium – An Additive Manufacturing Conference*, 1182–1200. PDF
5. **Shanto, T. A.**, Pavel, H. R., Ahmed, R., Abdullah, M., & Taylor, R. M. (2025). Leveraging large language models for process parameter optimization in 3D-printed ABS polymer specimens. *IISE Annual Conference and Expo 2025, Conference Proceedings*, 1351–1356. doi:10.21872/2025IISE_6901

6. **Shanto, T. A.**, Shahriar, M. A., Ahmed, T., Zulqernine, M. J., & Taylor, R. M. (2025). Predicting mechanical strength in FDM-printed ABS parts with in-process annealing: A machine-learning approach. *IISE Annual Conference and Expo 2025, Conference Proceedings*, 1224–1229. doi:10.21872/2025IISE_6734
7. Abdullah, M., Ozay, D., Ahmed, S. M. T., **Shanto, T. A.**, & Sridhar, E. P. (2025). Genetic algorithms in order-picking route optimization: A review of advances and implications for logistics. *IISE Annual Conference and Expo 2025, Conference Proceedings*, 1375–1380. doi:10.21872/2025IISE_6932
8. Ahmed, R., Niloy, R. S., Mozumder, M. R., & **Shanto, T. A.** (2024). Application of fused filament fabrication in the marine sector: From rapid prototyping to final product. *Proceedings of MARTEC 2024*. PDF
9. Zulqernine, M. J., Alam, M. A., Uddin, M. R., Dola, I. S., & **Shanto, T. A.** (2024). An investigation on the applications of additive manufacturing in the marine industry. *Proceedings of MARTEC 2024*. PDF

Patents

1. Taylor, R. M., Rane, R., Mrinal, M., Patel, P. T., & **Shanto, T. A.** *Fused filament fabrication print-head system*. U.S. Patent 12,496,776 B2. Patent record
2. **Shanto, T. A.**, Taylor, R. M., Mrinal, M., Ahmed, R., Patel, P. T., Rahman, M. M., Dola, I. S., & Deshpande, R. *Fused filament fabrication adaptive annealing print-head system*. U.S. Patent Application No. 19/660,770.

Teaching and Mentoring Experience

Teaching Assistant

The University of Texas at Arlington

Sep 2021–Present

Arlington, Texas, USA

- **Additive Manufacturing (MAE 5329; 200+ students):** Conducted laboratory exercises in SLA, SLS, material jetting, and FFF; trained students on universal testing, Simplify3D, PolyWorks, 3-Matic, and FARO arm scanning; mentored senior-design teams through 3D-printed UAV design, fabrication, testing, and competition activities.
- **Solid Mechanics (MAE 2312; 30 students):** Led supplemental problem-solving sessions, supported coursework and projects, held office hours, and evaluated assignments, quizzes, and examinations.
- **Additional courses:** Problems in MAE (MAE 1140; 100 students), Introduction to Manufacturing (MAE 3344; 50 students), Thermodynamics I (MAE 3310; 40 students), and Engineering Analysis (MAE 5332; 48 students); supported instruction, assessment, office hours, and project guidance.
- Selected for the I-ENGAGE Student Mentorship Program to guide undergraduate researchers in technical concepts, research methods, and problem-solving.

Professional and Industrial Experience

Co-Founder and Technical Lead

Increscent, Inc.

Aug 2022–Present

Texas, USA

- Translated a university-developed extrusion-based 3D-printing method into a staged R&D and commercialization roadmap with measurable performance milestones and qualification criteria.
- Authored technical proposals and grant applications, including a successful \$50,000 NSF I-Corps Team award and two SBIR Phase I proposals on AI-enabled extrusion-based additive manufacturing.
- Led structured customer discovery and translated voice-of-customer evidence into product requirements; prepared technical white papers, test reports, pitch decks, and partner/investor materials.

Mechanical Engineer

Super Petrochemical Limited

Oct 2018–Oct 2019

Bangladesh

- Designed process equipment and piping layouts in SolidWorks and AutoCAD Plant 3D, performed piping-stress analysis in CAESAR II, and supported proposal review for a \$10M mooring and pipeline program.
- Coordinated a 70-person mechanical workshop and maintenance crew, prioritized work orders, allocated resources, and enforced safe, high-quality repairs using Lean Six Sigma practices.
- Reduced oil-tanker construction and maintenance costs by approximately 10% through process-waste identification, planning improvements, and vendor coordination.

Honors, Grants, and Awards

- **NSF I-Corps Team Award, \$50,000:** High-Performance 3D-Printed Plastic Parts Through Novel Process Enhancement.
- **Maverick Business Pitch Competition, \$35,000 venture award:** Recognized for technical innovation, commercialization strategy, and business presentation.
- **Wheelwright Foundation Scholarship, \$2,000 (2023).**

- **I-ENGAGE Student Mentorship Program, \$2,000 stipend:** Selected to mentor undergraduate researchers.
- **Blackstone LaunchPad Campus Award, \$1,000 (2022).**

Professional Membership

American Society of Mechanical Engineers (ASME).

Academic References

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Professor in Practice

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